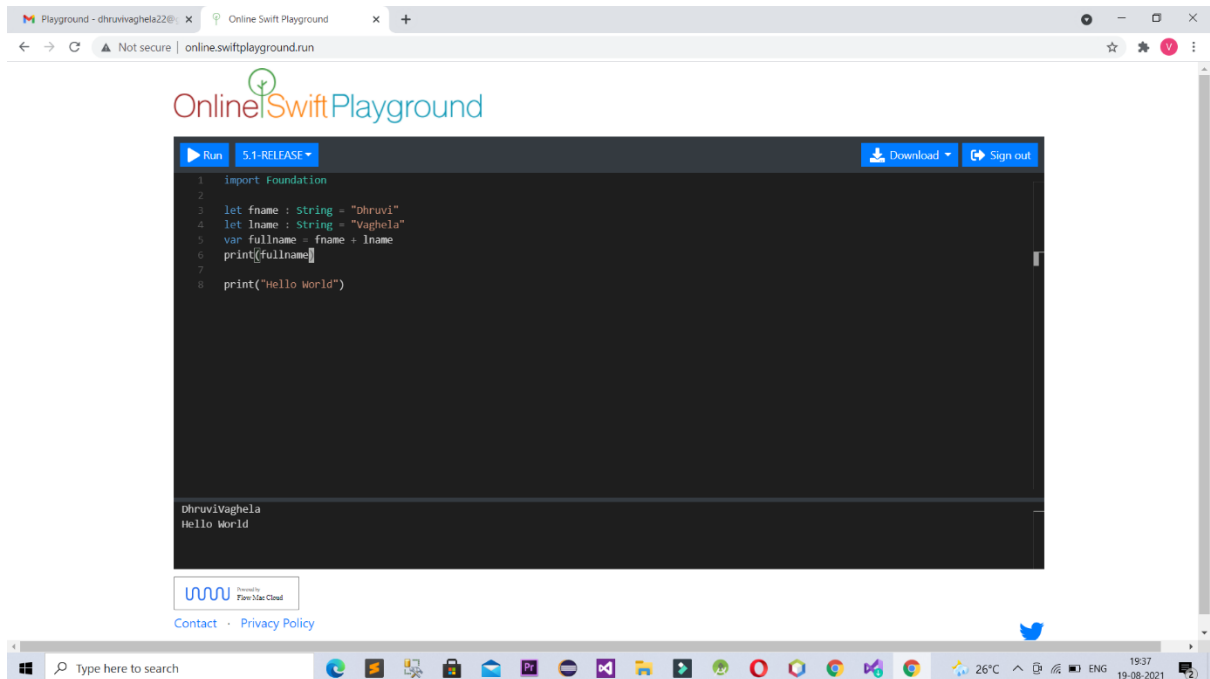


1. You are given the firstName and lastName of a user. Create a string variable called fullName that contains the full name of the user.



The screenshot shows a web browser window with the URL `online.swiftplayground.run`. The page title is "Online Swift Playground". The code editor contains the following Swift code:

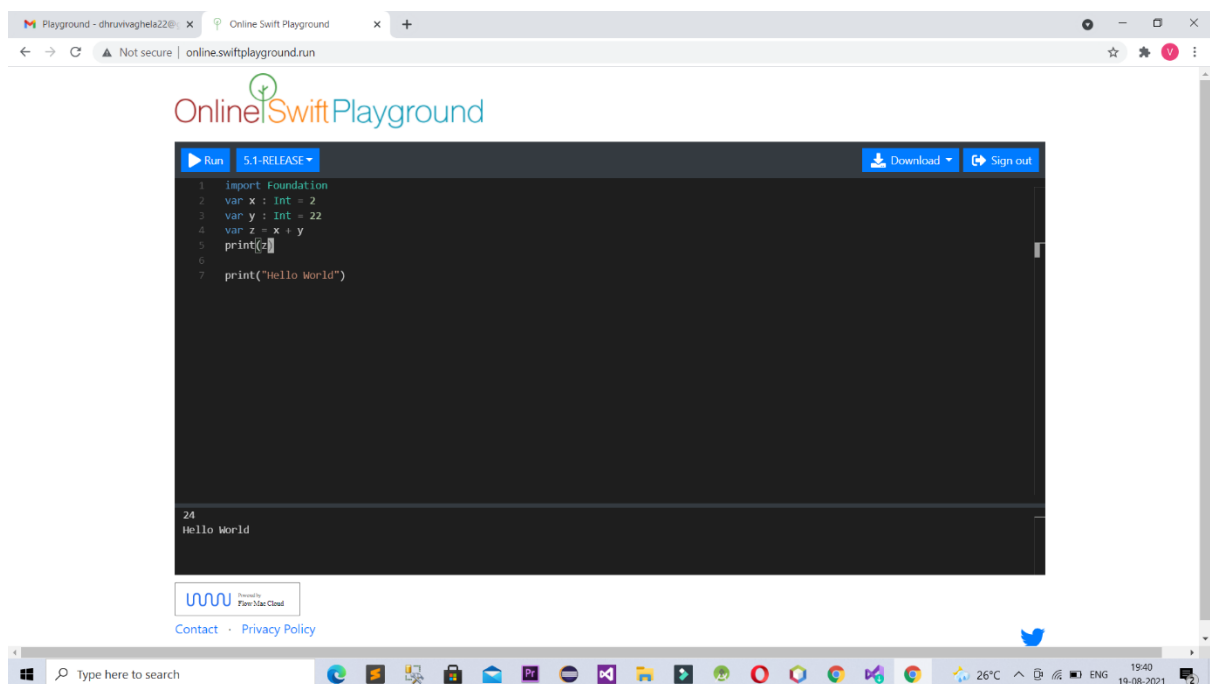
```
1 import Foundation
2
3 let fname : String = "Dhruvi"
4 let lname : String = "Vaghela"
5 var fullname = fname + lname
6 print(fullname)
7
8 print("Hello World")
```

The output console shows the results of the code execution:

```
DhruviVaghela
Hello World
```

At the bottom of the browser window, the Windows taskbar is visible, showing the search bar, task view button, and various application icons. The system tray shows the temperature as 26°C, the time as 19:37, and the date as 19-08-2021.

2. You are given two numbers a and b. Compute the sum of a and b and create a string stored in a variable named `formattedSum` that contains the sum written like bellow:

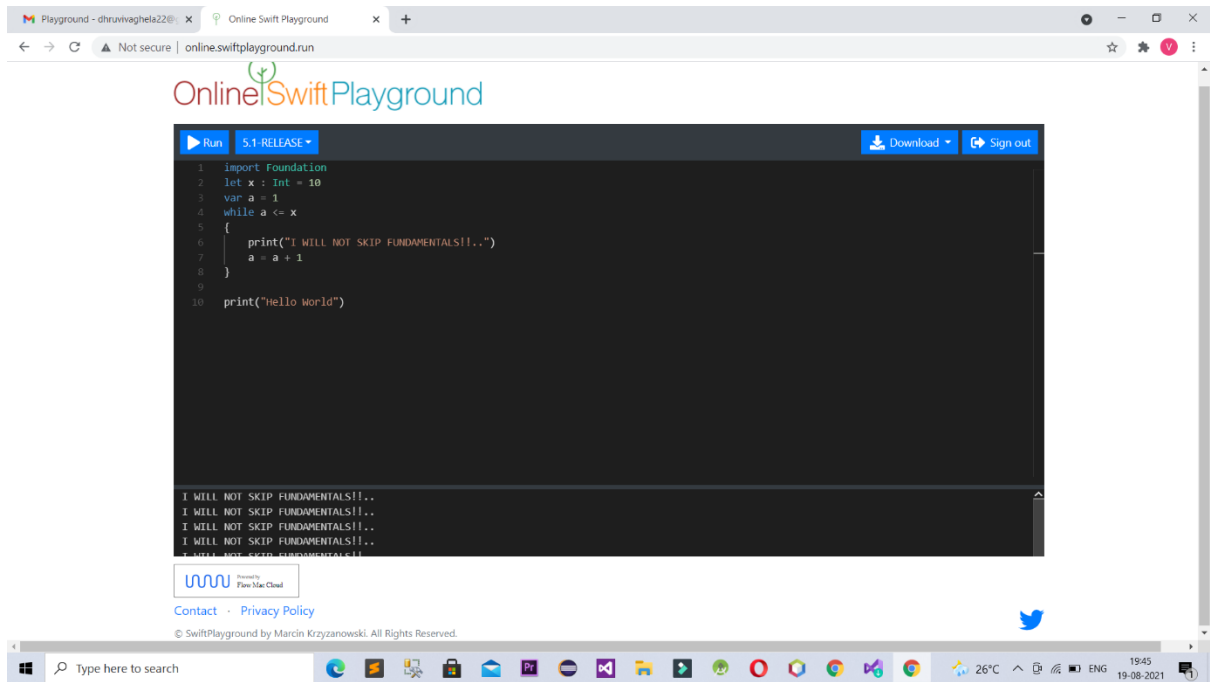


The screenshot shows the Online Swift Playground web interface. The browser address bar displays 'online.swiftplayground.run'. The page features the 'OnlineSwiftPlayground' logo and a code editor with the following Swift code:

```
1 import Foundation
2 var x : Int = 2
3 var y : Int = 22
4 var z = x + y
5 print(z)
6
7 print("Hello World")
```

Below the code editor, the output is displayed as '24' and 'Hello World'. The interface includes buttons for 'Run', 'Download', and 'Sign out'. At the bottom, there is a 'Powered by Firebase Cloud' logo and links for 'Contact' and 'Privacy Policy'. The Windows taskbar at the bottom shows the search bar, various application icons, and the system tray with a temperature of 26°C and the date 19-08-2021.

3. Write a program that writes “I will not skip the fundamentals!” N times



The screenshot shows a web browser window with the URL `online.swiftplayground.run`. The page title is "Online Swift Playground". The code editor contains the following Swift code:

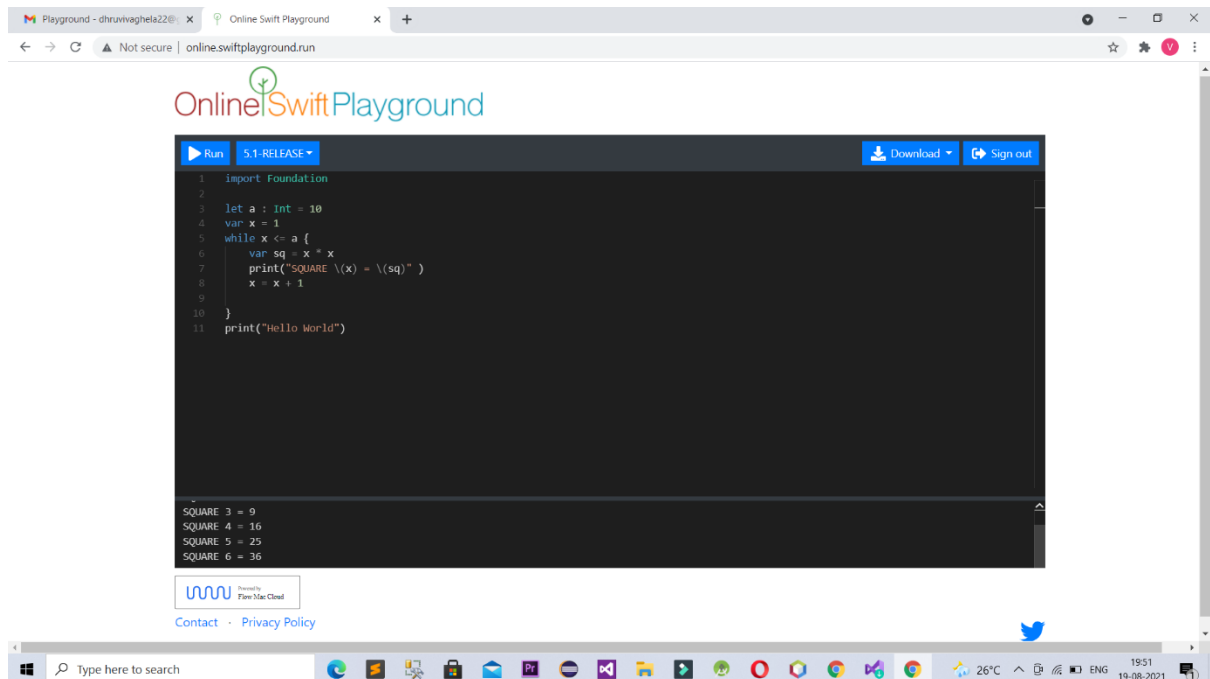
```
1 import Foundation
2 let x : Int = 10
3 var a = 1
4 while a <= x
5 {
6     print("I WILL NOT SKIP FUNDAMENTALS!!..")
7     a = a + 1
8 }
9
10 print("Hello World")
```

The output area shows the following text:

```
I WILL NOT SKIP FUNDAMENTALS!!..
I WILL NOT SKIP FUNDAMENTALS!!..
I WILL NOT SKIP FUNDAMENTALS!!..
I WILL NOT SKIP FUNDAMENTALS!!..
I WILL NOT SKIP FUNDAMENTALS!!..
```

The browser's taskbar at the bottom shows various application icons, including Microsoft Edge, File Explorer, and several other utilities. The system tray on the right indicates a temperature of 26°C, signal strength, and the date and time: 19:45 on 19-08-2021.

4. Print the first N square numbers.



The screenshot shows a web browser window with the URL `online.swiftplayground.run`. The page title is "Online Swift Playground". The code editor contains the following Swift code:

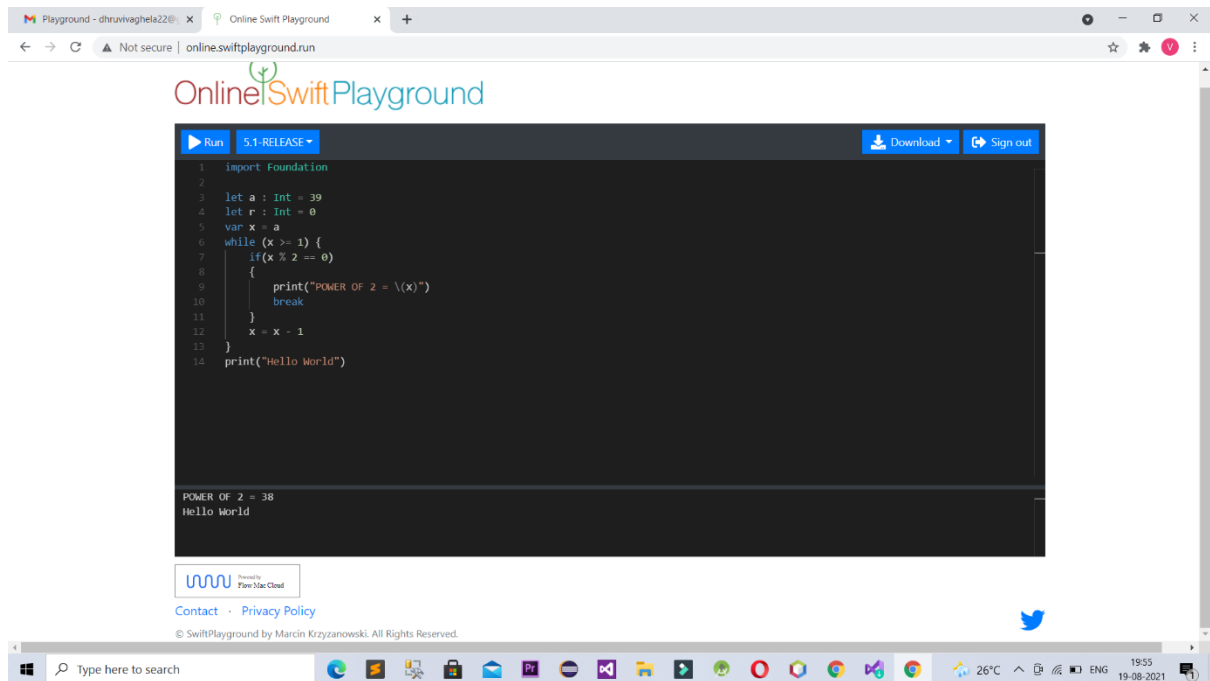
```
1 import Foundation
2
3 let a : Int = 10
4 var x = 1
5 while x <= a {
6     var sq = x * x
7     print("SQUARE \(x) = \(sq)")
8     x = x + 1
9 }
10
11 print("Hello World")
```

The output console shows the following results:

```
SQUARE 3 = 9
SQUARE 4 = 16
SQUARE 5 = 25
SQUARE 6 = 36
```

The browser's taskbar at the bottom shows the Windows logo, a search bar, and various application icons. The system tray on the right shows the temperature as 26°C, the time as 19:51, and the date as 19-08-2021.

5. Print the powers of 2 that are less than or equal to N.



The screenshot shows a web browser window with the URL `online.swiftplayground.run`. The page title is "Online Swift Playground". The code editor contains the following Swift code:

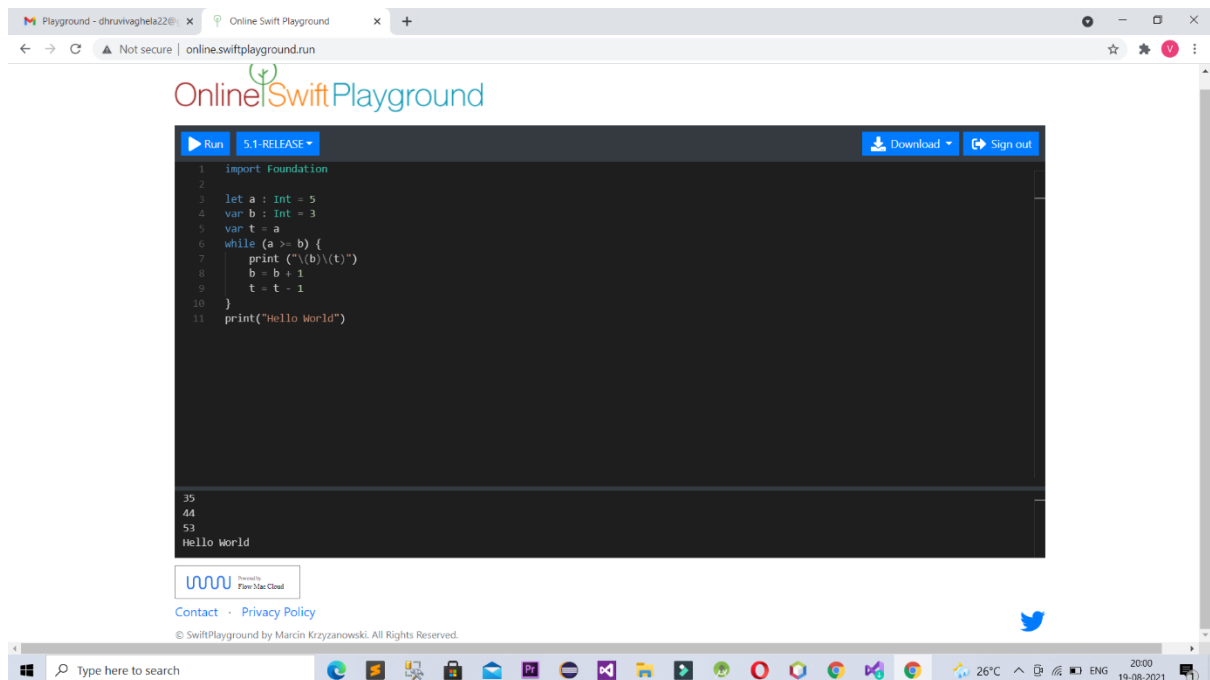
```
1 import Foundation
2
3 let a : Int = 39
4 let r : Int = 0
5 var x = a
6 while (x >= 1) {
7     if(x % 2 == 0)
8     {
9         print("POWER OF 2 = \(x)")
10        break
11    }
12    x = x - 1
13 }
14 print("Hello World")
```

The output of the code is displayed below the editor:

```
POWER OF 2 = 38
Hello World
```

The browser window also shows a "Run" button, a "5.1-RELEASE" dropdown, a "Download" button, and a "Sign out" button. The footer of the playground includes a "Contact" link, a "Privacy Policy" link, and a copyright notice: "© SwiftPlayground by Marcin Krzyzanowski. All Rights Reserved."

6. Write all the numbers from 1 to N in alternative order, one number from the left side (starting with one) and one number from the right side (starting from N down to 1).



The screenshot shows a web browser window with the URL `online.swiftplayground.run`. The page title is "Online Swift Playground". The code editor contains the following Swift code:

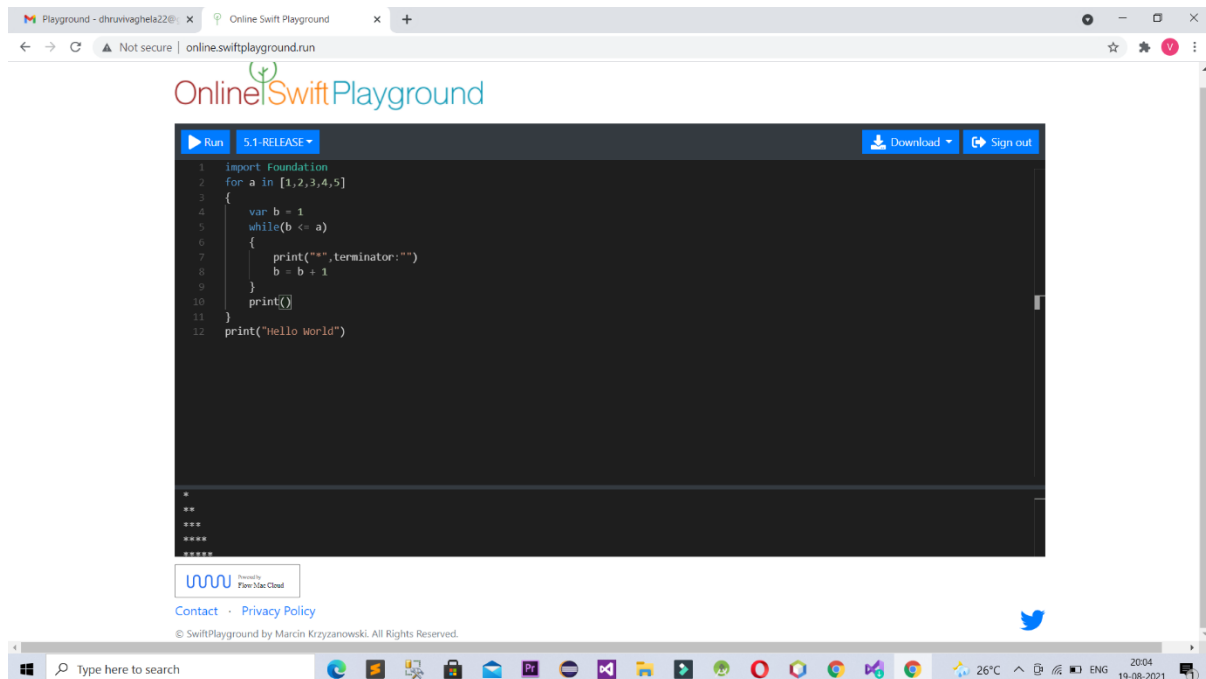
```
1 import Foundation
2
3 let a : Int = 5
4 var b : Int = 3
5 var t = a
6 while (a >= b) {
7     print("\(b)\(t)")
8     b = b + 1
9     t = t - 1
10 }
11 print("Hello World")
```

The output area shows the following text:

```
35
44
53
Hello World
```

At the bottom of the page, there is a footer with the text: "© SwiftPlayground by Marcin Krzyzanowski. All Rights Reserved."

7. Given an integer N draw a triangle of asterisks. The triangle should have N lines, the i-th line should have i asterisks on it.



The screenshot shows a web browser window with the URL `online.swiftplayground.run`. The page title is "Online Swift Playground". The code editor contains the following Swift code:

```
1 import Foundation
2 for a in [1,2,3,4,5]
3 {
4     var b = 1
5     while(b <= a)
6     {
7         print(" ", terminator: "")
8         b = b + 1
9     }
10    print()
11 }
12 print("Hello World")
```

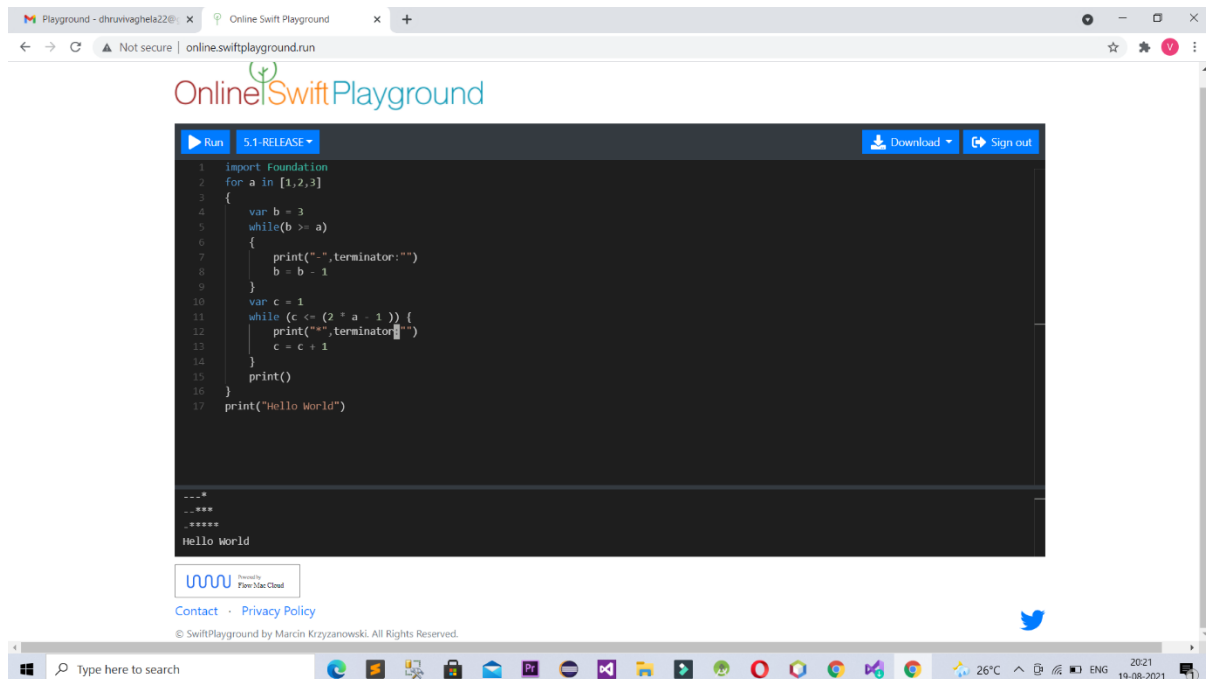
The output area shows the result of the code execution:

```

*
**
***
****
*****
Hello World
```

The browser's taskbar at the bottom shows the Windows logo, a search bar, and various application icons. The system tray on the right shows the temperature (26°C), battery level, and the date and time (2004 19-08-2021).

8. Given an integer N draw a pyramid of asterisks. The pyramid should have N lines. On the i-th line there should be N-i spaces followed by $i*2-1$ asterisks.



The screenshot shows the Online Swift Playground interface. The code editor contains the following Swift code:

```
1 import Foundation
2 for a in [1,2,3]
3 {
4     var b = 3
5     while(b >= a)
6     {
7         print("-",terminator:"")
8         b = b - 1
9     }
10    var c = 1
11    while (c <= (2 * a - 1)) {
12        print("+",terminator:"")
13        c = c + 1
14    }
15    print()
16 }
17 print("Hello World")
```

The output area shows the following text:

```
---*
..***
*****
Hello world
```

The interface includes a "Run" button, a "5.1-RELEASE" dropdown, a "Download" button, and a "Sign out" button. The footer of the playground shows "© SwiftPlayground by Marcin Krzyzanowski. All Rights Reserved." and a Twitter icon.